

An update to the sardine assessment using data from 1984-2024, with associated implications for 2025 management recommendations

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The S_{base} sardine assessment of de Moor (2025a,b) has been updated to fit to data available up to March 2025. This model indicates a small decrease in the sardine biomass off the West Coast over the past year and a substantial decrease in the sardine biomass off the South Coast.

Keywords: assessment, biomass, hydroacoustic survey, sardine, South Africa

Background

Following the draft results from the recent small pelagic biomass hydroacoustic survey (Coetzee *et al.* 2025), questions were raised in the recent SWG-PEL meeting as to how the biomass of sardine off South Africa's south coast apparently decreased substantially within only one year. This question related to a change in the survey estimate of total biomass. This document updates the stock assessment for South African sardine from the end of 2023 to the end of 2024 to include the most recent data to investigate what a combination of all the data – not only the (delayed) November surveys – suggest has recently occurred with sardine recruitment and total biomass on each coast.

The current estimated status of the sardine resource provides direct input into the management recommendations for 2025 because Exceptional Circumstances have been declared for sardine since December 2018 and management recommendations for the sardine fishery have thus been based on alternative *ad hoc* methods, rather than OMP recommendations.

Data

The assessment is fitted to the 1984-2023 data as detailed in de Moor *et al.* 2024, in addition to 2024 data as detailed in the Appendix.

Methods

The model used is S_{base} from de Moor (2025a,b), except for the extension to the end of $y_n = 2024$. The 2024 survey is modelled to have taken place at 1 April 2025, i.e. equation (A9) of de Moor (2025a) is extended to:

$$NLmid_{c,p,y,l}^j = \begin{cases} \sum_{a=0}^{5+} A_{c,y,a,l} (Nmid_{c,p,y,4,a}^j - C_{c,p,y,4,a}^j) & y_0 \leq y \leq 2022 \\ \sum_{a=0}^{5+} A_{c,y+1,a,1,l} \left((Nmid_{c,p,y,4,a}^j - C_{c,p,y,4,a}^j) e^{-M_{w,y+1,a}^j/4} - C_{c,p,y+1,1,a}^j \right) e^{-M_{w,y+1,a}^j/24} & y = 2023 \\ \sum_{a=0}^{5+} A_{c,y+1,a,1,l} \left((Nmid_{c,p,y,4,a}^j - C_{c,p,y,4,a}^j) e^{-M_{w,y+1,a}^j/4} - C_{c,p,y+1,1,a}^j \right) e^{-M_{w,y+1,a}^j/8} & y = y_n \end{cases}$$

In addition, given there is little data to reliably estimate recruitment in the final year, y_n , the uninformative prior distribution is replaced with $R_{yn}^j \sim N(\mu_j, \sigma_j^2)$, where μ_j is the average model estimated recruitment between y_{n-5} and y_{n-1} for stock j and σ_j^2 is the associated variance.

Results and Discussion

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Results are shown in Figures 1 to 13. This model estimates a small increase of sardine biomass off the west coast over the past year and a substantial decrease in biomass off the south coast. Recruitment in the final year is now influenced by the informative prior. Recruitment of WTS to the west coast is estimated to be low in recent years (Figure 10) and <40% of WTS recruits and adults were estimated to actively move from the west to the south coast in 2024. This is the lowest proportion since 2011.

Acknowledgements

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References

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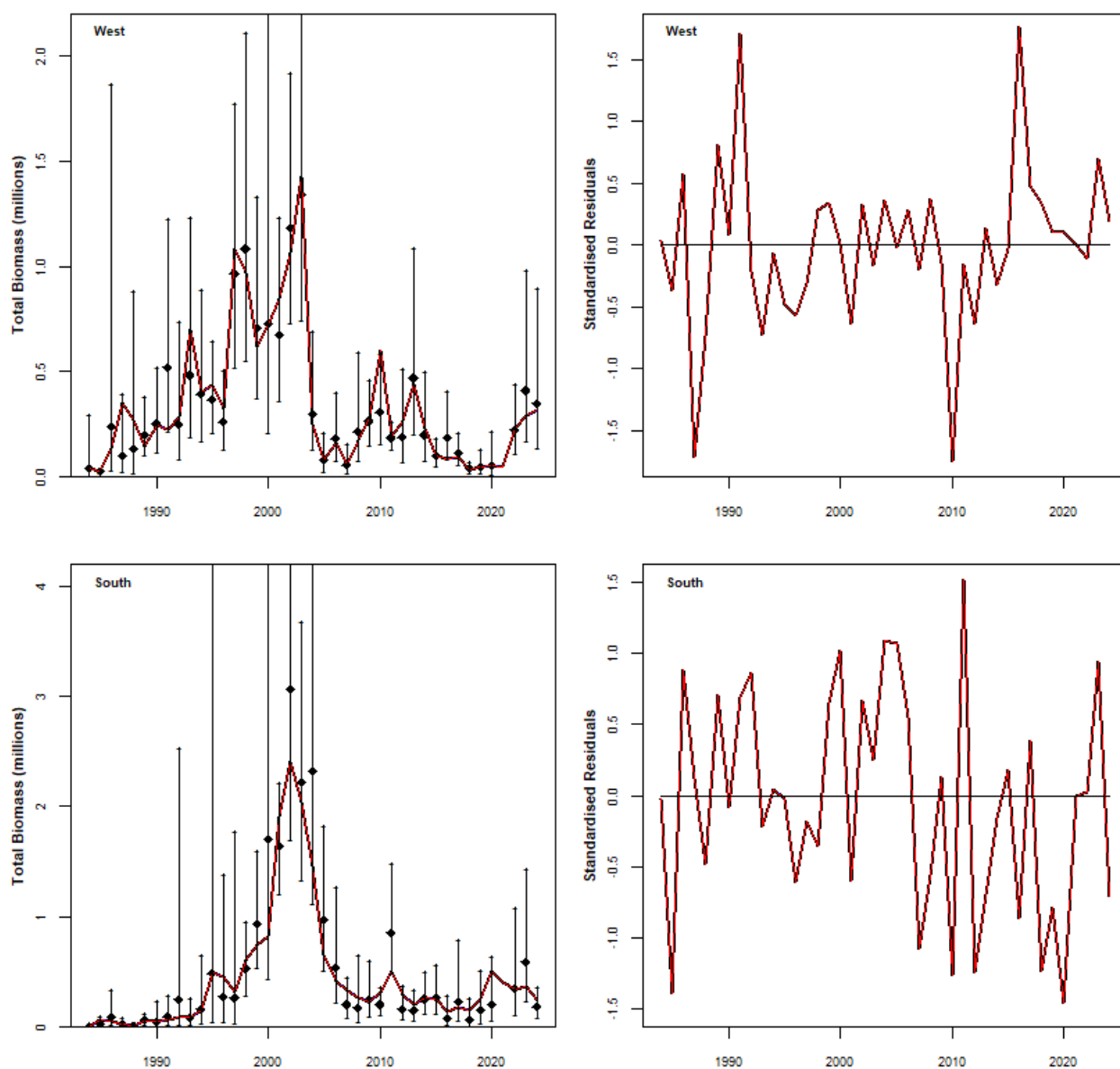


Figure 1. Acoustic survey estimated and model predicted November sardine total biomass from 1984 to 2024. The observed indices are shown with 95% confidence intervals. The standardised residuals (i.e. the residual divided by the corresponding standard deviation, including additional variance where appropriate) from the fits are given in the right hand plots.

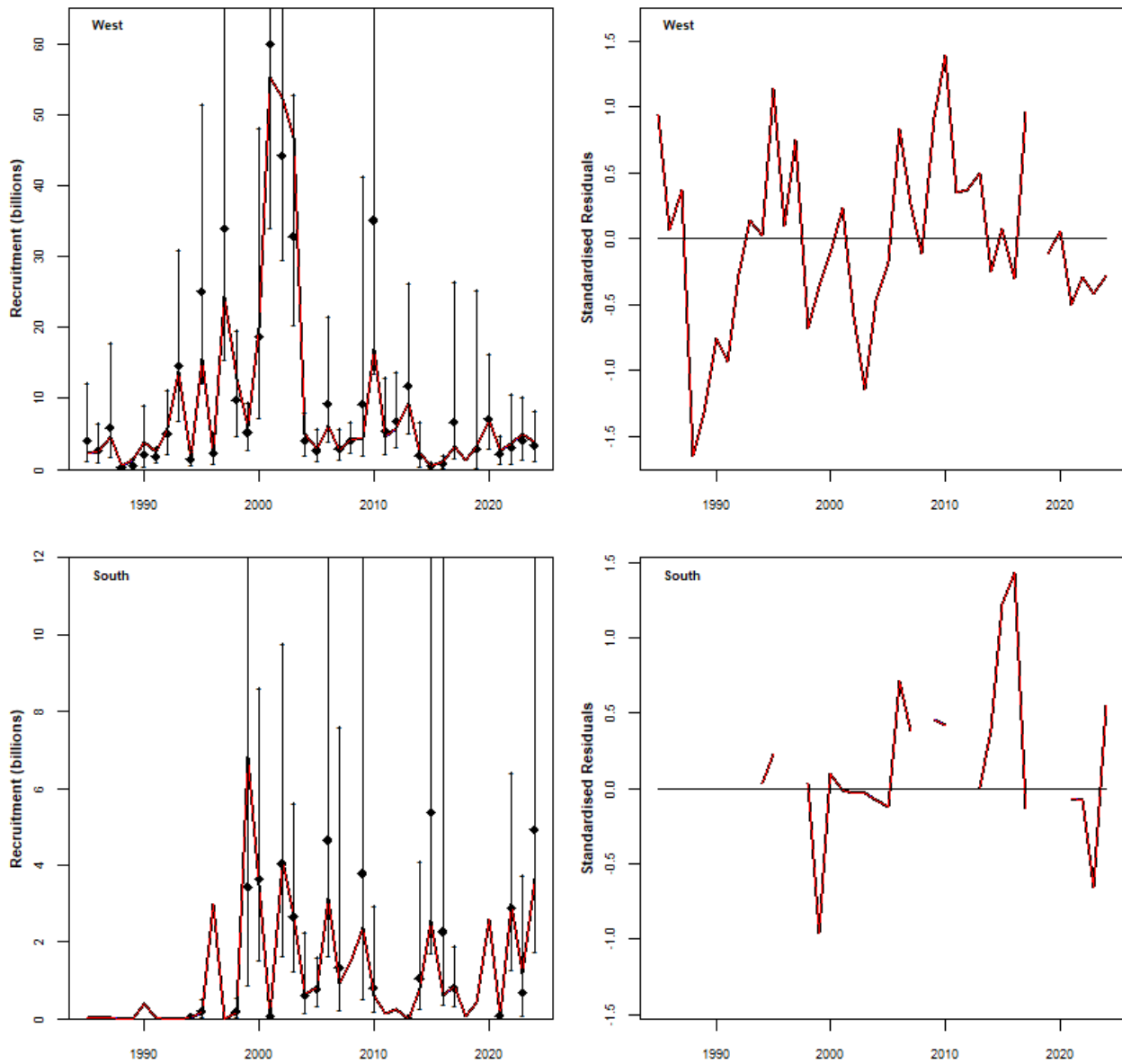


Figure 2. Acoustic survey estimated and model predicted sardine recruitment numbers from May/June 1985 to 2024. There was no survey observation in 2018; the model predicted value corresponds to the recruitment predicted at 8th June 2018 which is the average start date of the survey from 2016, 2017 and 2019 surveys. The survey indices are shown with 95% confidence intervals. The standardised residuals from the fit are given in the right hand plots.

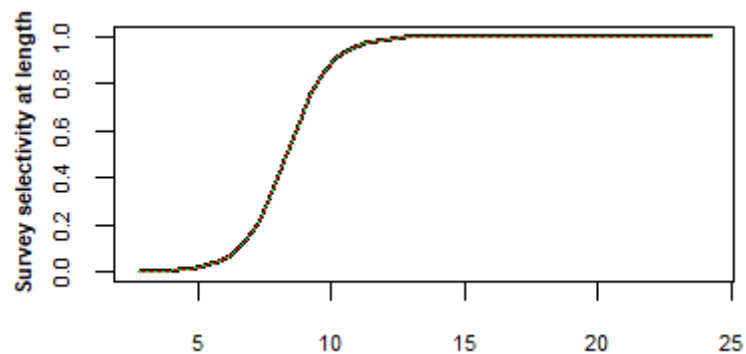


Figure 3. The model estimated November survey selectivity at length.

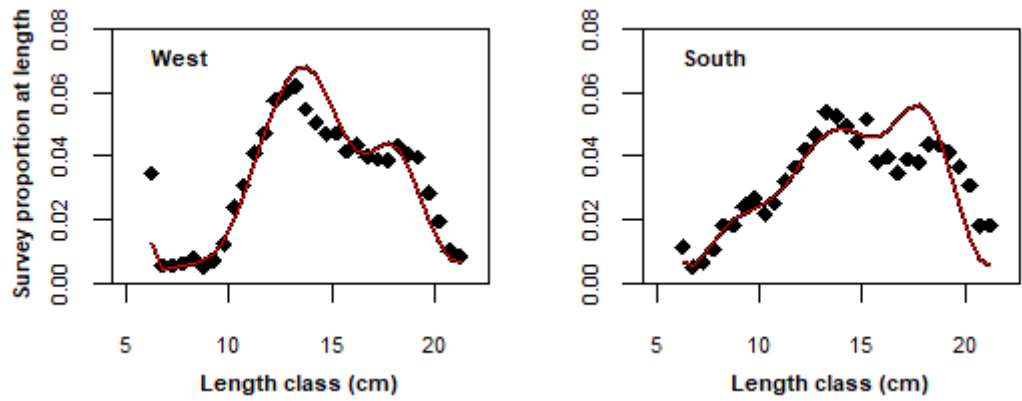


Figure 4. Average (over all years) model predicted and observed proportion-at-length in the November survey.

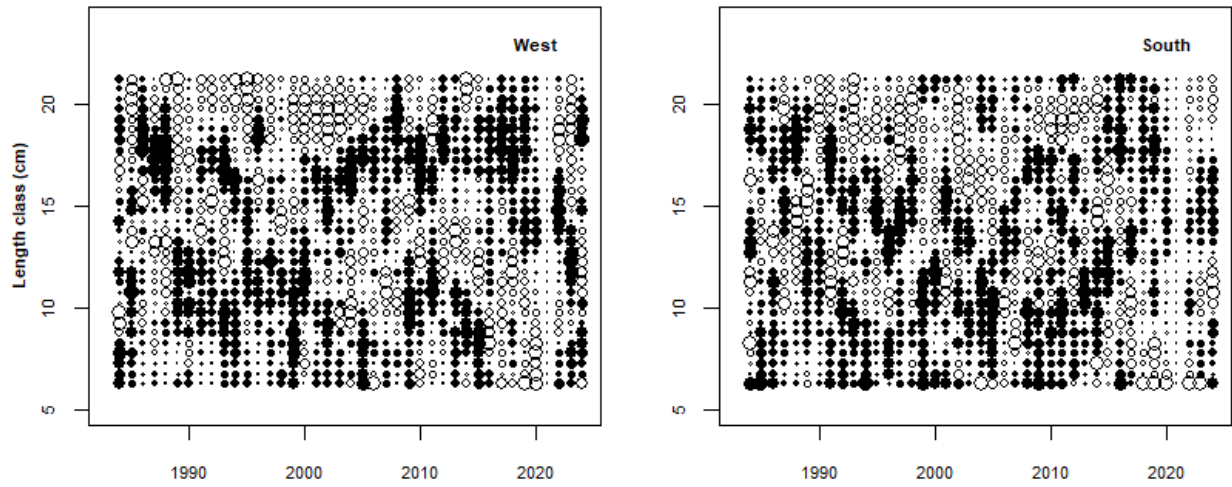


Figure 5. Residuals from the fit of the model predicted proportions-at-length in the November survey to the hydroacoustic survey estimated proportions.

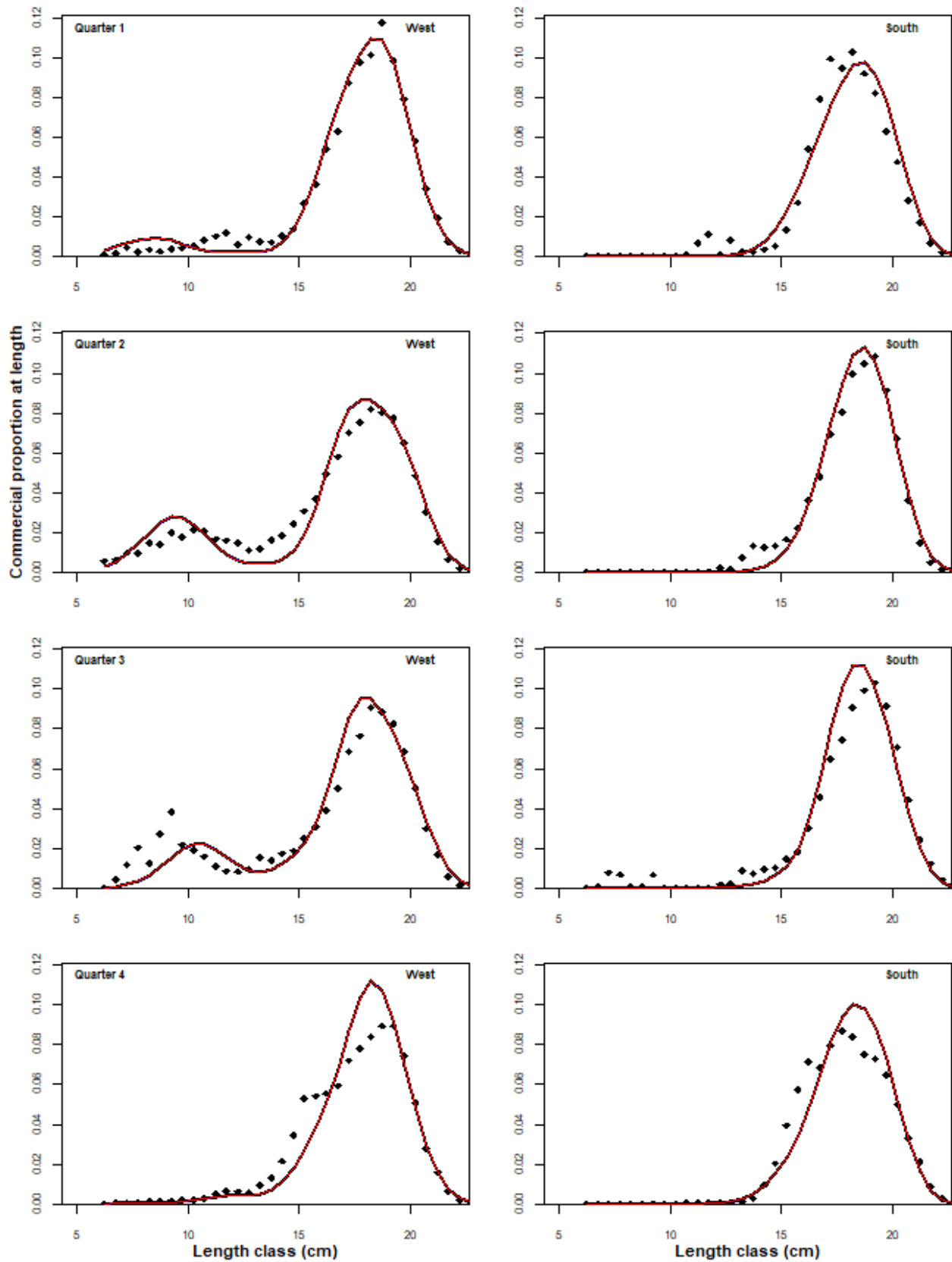


Figure 6. Average (over all years) quarterly model predicted and observed proportions-at-length in the commercial catch.

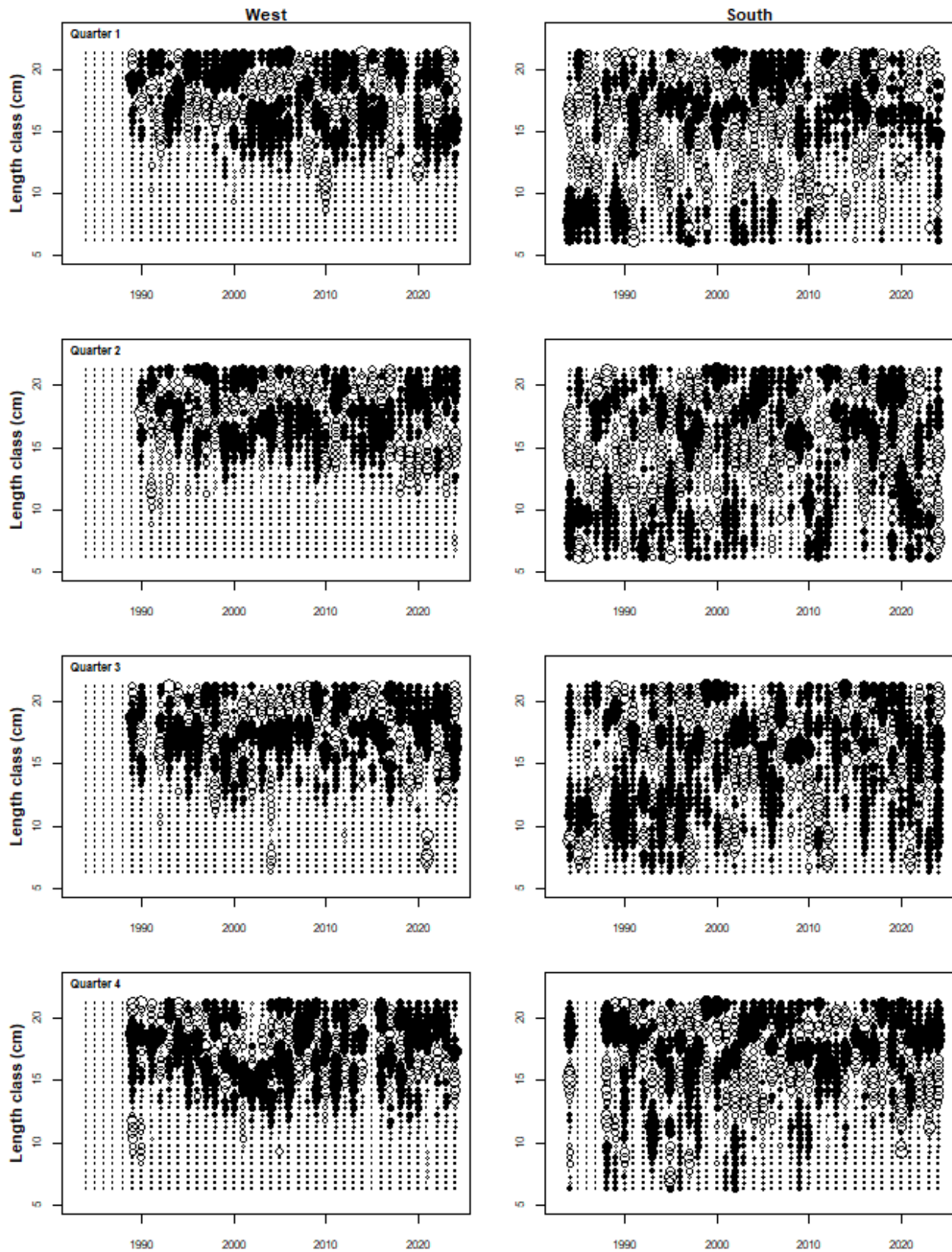


Figure 7. Residuals from the fit of the model predicted proportions-at-length in the quarterly commercial catch to the observed proportions.

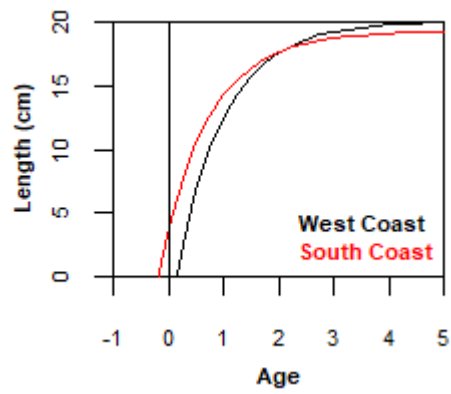


Figure 8. The von Bertalanffy growth curves. (No differences in co-horts assumed).

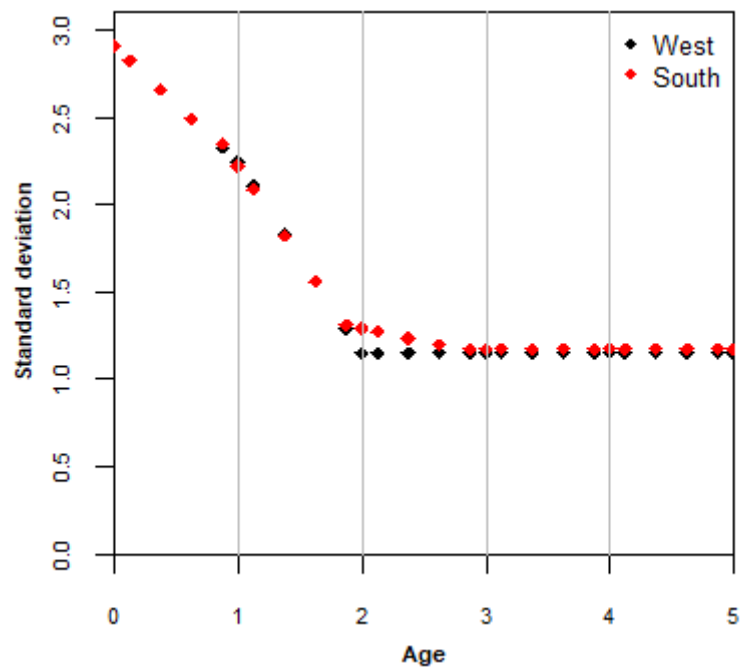


Figure 9. The estimated standard deviation about the mean length-at-age.

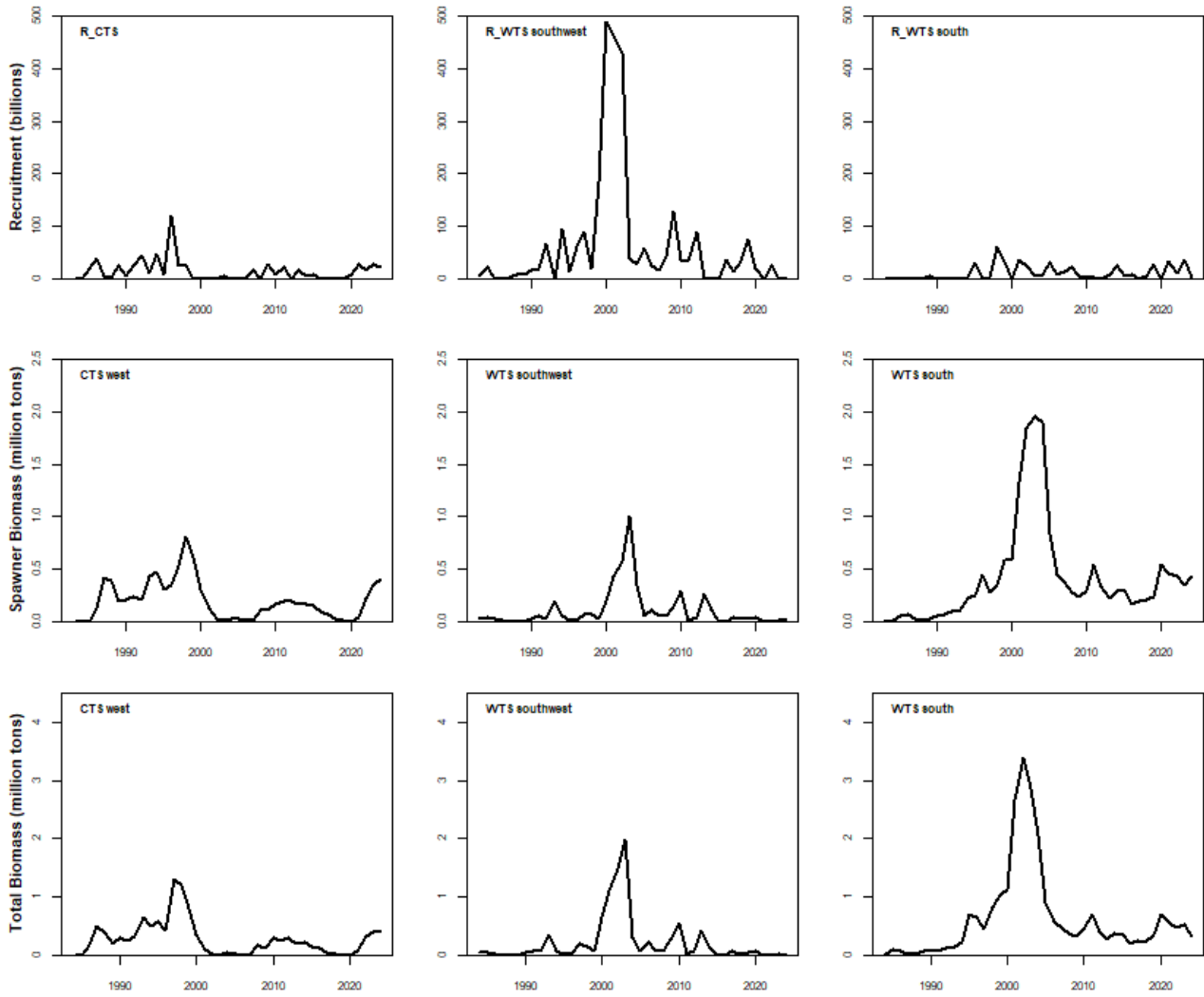


Figure 10. Model predicted sardine recruitment (top row), spawner biomass (2nd row) and total biomass (lower row) from November 1983 to November 2024.

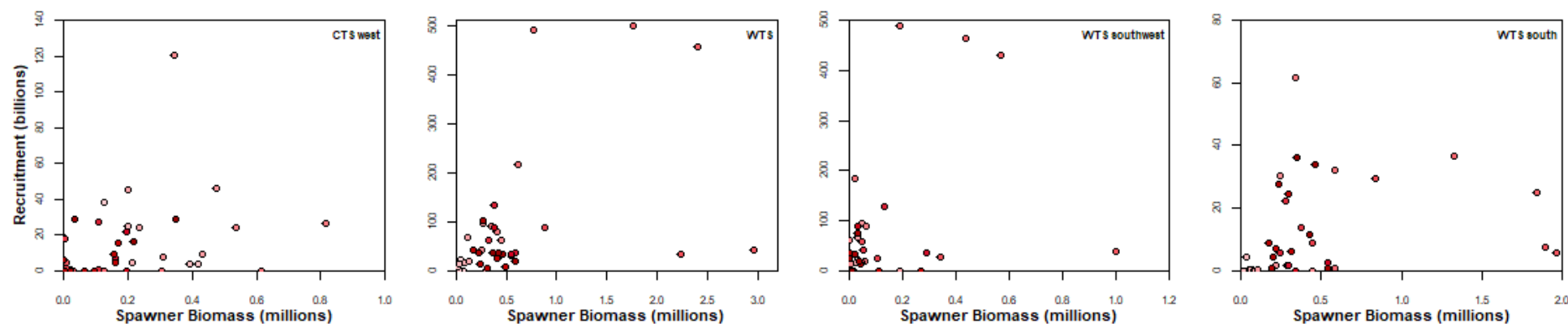


Figure 11. Model predicted sardine recruitment plotted against spawner biomass, with colouring indicating earlier (lighter) to more recent (darker) years. Stock recruitment relationships might be considered for the left two columns. The right two columns show the resultant WTS recruitment by coast, which is due both to spawning by coast and passive transport of spawning products from south coast spawning to west coast recruitment.

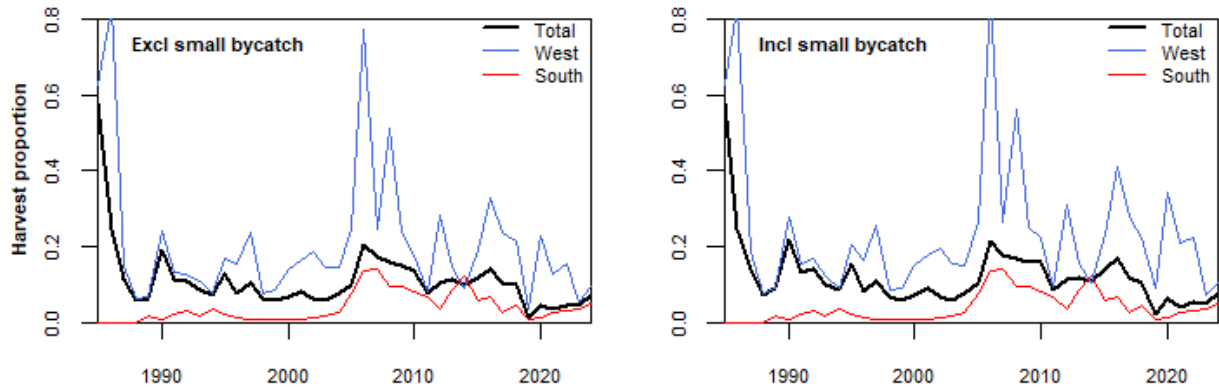


Figure 12. Historical exploitation rates by coast, calculated as the annual (January – December) catch tonnage as a proportion of the November biomass of the previous year. Exploitation rates are shown excluding (left) and including (right) the small sardine bycatch.

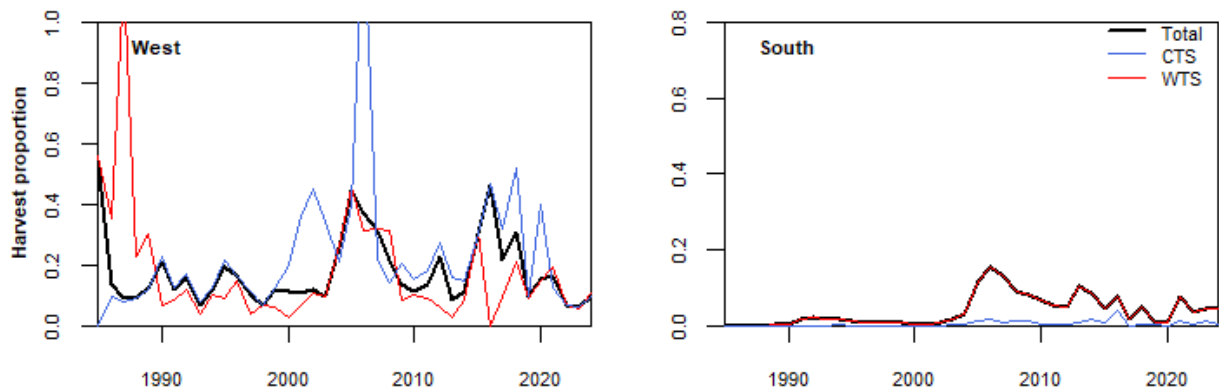


Figure 13. Historical exploitation rates by stock and coast calculated as the model predicted catch tonnage (numbers multiplied by weight) by stock and coast (summed over the four quarters from January to December) as a proportion of the average quarterly stock- and coast-disaggregated total biomass. This can result in proportions >1, particularly when biomass decreases substantially between quarters within a year.

Appendix: Sardine data for 2024

The following data are available for 2024 (Merkle *et al.* 2024, Coetzee *et al.* 2025, Merkle pers comm):

- May 2024 recruitment survey began at 1st June, i.e. $t_y = 1.067$.
- May 2024 recruit cut-off length was <11.5cm.
- May 2024 recruit numbers and inter-transect CVs were 3.365 billion (0.466) and 4.931 billion (0.524) for west and east of Cape Agulhas, respectively.

- November 2024 total biomass survey took place over March-April 2025 and is modelled to have occurred on 1 April 2025 instead of 15th November 2024.
- The total biomass and inter-transect CVs were 322 756t (0.490) and 164 873t (0.348) for west and east of Cape Agulhas, respectively.
- The March-April 2025 survey length frequencies for west and east of Cape Agulhas.

- Commercial catch tonnages and length frequencies from January 2024 to March 2025, for west of Cape Point, Cape Point to Cape Agulhas and east of Cape Agulhas.
- (By)Catch tonnage is available for all January – March months, areas and size classes in 2025. However, length frequency data are not available for all months/areas in 2025:
- For the ≤ 14 cm sardine bycatch in January – March 2025, the weighted average relative length frequency from 2022-2024 in the same months/areas are used.
- For the >14 cm sardine bycatch with round herring and anchovy in March 2025, the corresponding relative length frequency from the >14 cm directed sardine in March 2025 is used.
- Catch from 1-2 June 2025 (to contribute to recruit catch from 1 April to the day before the survey) consisted only of ≤ 14 cm and >14 cm bycatch in the area north of Cape Point.
- As maturity data could not be collected during November 2024, the average 2017-2020,2022 $L_{50,y}$ and δ_y^{mat} were used for November 2024, as assumed for November 2023 (de Moor *et al.* 2025a).